

API (cricketdata.org)

This Python script defines and then calls a function named `getLiveScores`. The function's purpose is to fetch live cricket scores from an API (a web service) and print the details for each match it finds.

Here is the breakdown of its execution:

1. `with open('api.txt', 'r') as file: ... apiUrl = file.read()`
 - This block opens a local file named `api.txt` in read mode (`'r'`).
 - `file.read()` reads the *entire* content of the file into the `apiUrl` variable.
 - This is good practice for storing sensitive data like API keys or URLs, so they aren't hard-coded directly in the script.
2. `response = requests.get(apiUrl)`
 - This is the core of the script. It uses the `requests` library to send an HTTP `GET` request to the URL that was read from your `api.txt` file.
 - The server's reply (including data, status codes, and headers) is stored in the `response` variable.
3. `if response.status_code == 200:`
 - This checks if the request was successful. A status code of `200` means "OK."
 - If the code is *not* `200` (e.g., `404 Not Found`, `401 Unauthorized`), it skips this block and jumps to the `else` statement (see step 13).
4. `data = response.json()`
 - Assuming the request was successful, this line takes the raw JSON data from the response (which is just text) and parses it into a native Python dictionary called `data`. This makes it easy to access data using keys (e.g., `data['key']`).
5. `if data.get("status") == "success" and "data" in data and data["data"]`
 - This is a *second* level of validation. Just because the HTTP request was `200 OK` doesn't mean the API *found* any data.
 - This `if` statement checks the *content* of the data dictionary to ensure:
 - `data.get("status") == "success"`: The API's own internal status is "success."
 - `"data" in data`: A key named "data" actually exists.
 - `data["data"]`: The "data" key's value is not empty (an empty list `[]` would fail this check).
6. `matchList = data["data"]`
 - If the data is valid, this line assigns the list of matches (which is the value of the "data" key) to a new, more readable variable `matchList`.
7. `for match in matchList:`
 - This line starts a `for` loop, which will iterate through every item (each a `match` dictionary) inside the `matchList`.
8. `print("-" * 70) ... print(f"Date: {match['date']}")`
 - Inside the loop, these `print` statements access the values from the `match` dictionary (using keys like `'name'`, `'status'`, etc.) and print them to the console in a formatted way. The `---` line just acts as a visual separator.
9. `except requests.exceptions.RequestException as e:`
 - This is the error-handler for the `try` block. If the `requests.get()` call fails completely (e.g., no internet, DNS error), this code will catch that `RequestException`, print a clear error message, and prevent the program from crashing.

```
In [19]: import requests

def getLiveScores():
    print("Fetching live cricket scores...")
    with open('api.txt', 'r') as file:
        apiUrl = file.read()

    try:
        response = requests.get(apiUrl)

        if response.status_code == 200:
            # Parse the JSON response into a Python dictionary.
            data = response.json()

            # Check if the API call was successful and data exists.
            if data.get("status") == "success" and "data" in data and data["data"]:
                matchList = data["data"]
                print(f"\nFound {len(matchList)} matches.\n")

                # Loop through the list of matches and print the details.
                for match in matchList:
                    print("-" * 70)
                    print(f"Match: {match['name']}")
                    print(f"Status: {match['status']}")
                    print(f"Venue: {match['venue']}")
                    print(f>Date: {match['date']}")

            else:
                # Handle cases where the API returns an error message in the JSON
                errorMessage = data.get("reason", "No live matches found or an unknown error occurred.")
                print(f"Could not retrieve matches: {errorMessage}")

        else:
            # Handle HTTP errors (e.g., 401 Unauthorized, 404 Not Found)
            print(f"Error: Failed to fetch data. Status code: {response.status_code}")
            print("Response:", response.text)

    except requests.exceptions.RequestException as e:
```

```
# Handle network-related errors (e.g., no internet connection)
print(f"A network error occurred: {e}")
```

```
getLiveScores()
```

Fetching live cricket scores...

Found 9 matches.

Match: Pakistan vs South Africa, 1st T20I
Status: South Africa won by 55 runs
Venue: Rawalpindi Cricket Stadium, Rawalpindi
Date: 2025-10-28

Match: Zimbabwe vs Afghanistan, 1st T20I
Status: Afghanistan won by 53 runs
Venue: Harare Sports Club, Harare
Date: 2025-10-29

Match: Bangladesh vs West Indies, 2nd T20I
Status: West Indies won by 14 runs
Venue: Bir Sreshtho Flight Lieutenant Matiur Rahman Stadium, Chattogram
Date: 2025-10-29

Match: Western Australia vs South Australia, 9th Match
Status: Day 2: Stumps - South Australia trail by 86 runs
Venue: W.A.C.A. Ground, Perth
Date: 2025-10-28

Match: Queensland vs New South Wales, 8th Match
Status: Day 2: Stumps - Queensland opt to bowl
Venue: The Gabba, Brisbane
Date: 2025-10-28

Match: Victoria vs Tasmania, 7th Match
Status: Day 2: Stumps - Victoria lead by 59 runs
Venue: Junction Oval, Melbourne
Date: 2025-10-27

Match: New Zealand vs England, 2nd ODI
Status: New Zealand won by 5 wkts
Venue: Seddon Park, Hamilton
Date: 2025-10-29

Match: England Women vs South Africa Women, 1st Semi-Final
Status: South Africa Women won by 125 runs
Venue: Barsapara Cricket Stadium, Guwahati
Date: 2025-10-29

Match: Australia vs India, 1st T20I
Status: No result (due to rain)
Venue: Manuka Oval, Canberra
Date: 2025-10-29

Same code with `tabulate` module for better formatting

```
In [20]: import requests
from tabulate import tabulate

def getLiveScores():
    print("Fetching live cricket scores...")
    with open('api.txt', 'r') as file:
        apiUrl = file.read()

    try:
        response = requests.get(apiUrl)

        if response.status_code == 200:
            # Parse the JSON response into a Python dictionary.
            data = response.json()

            # Check if the API call was successful and data exists.
            if data.get("status") == "success" and "data" in data and data["data"]:
                matchList = data["data"]
                print(f"\nFound {len(matchList)} matches.\n")

                # Loop through the list of matches and print the details.
                matches = []
                for match in matchList:
                    matchDictionary = {
                        'Match': match['name'],
                        'Status': match['status'],
```

```
        'Venue': match['venue'],
        'Date': match['date']
    }
    matches.append(matchDictionary)

print(tabulate(matches, headers="keys", tablefmt="grid"))

else:
    # Handle cases where the API returns an error message in the JSON
    errorMessage = data.get("reason", "No live matches found or an unknown error occurred.")
    print(f"Could not retrieve matches: {errorMessage}")

else:
    # Handle HTTP errors (e.g., 401 Unauthorized, 404 Not Found)
    print(f"Error: Failed to fetch data. Status code: {response.status_code}")
    print("Response:", response.text)

except requests.exceptions.RequestException as e:
    # Handle network-related errors (e.g., no internet connection)
    print(f"A network error occurred: {e}")

getLiveScores()
```

Fetching live cricket scores...

Found 9 matches.

Match	Status	Venue	Date
Pakistan vs South Africa, 1st T20I	South Africa won by 55 runs	Rawalpindi Cricket Stadium, Rawalpindi	2025-10-28
Zimbabwe vs Afghanistan, 1st T20I	Afghanistan won by 53 runs	Harare Sports Club, Harare	2025-10-29
Bangladesh vs West Indies, 2nd T20I	West Indies won by 14 runs	Bir Sreshtho Flight Lieutenant Matiur Rahman Stadium, Chattogram	2025-10-29
Western Australia vs South Australia, 9th Match	Day 2: Stumps - South Australia trail by 86 runs	W.A.C.A. Ground, Perth	2025-10-28
Queensland vs New South Wales, 8th Match	Day 2: Stumps - Queensland opt to bowl	The Gabba, Brisbane	2025-10-28
Victoria vs Tasmania, 7th Match	Day 2: Stumps - Victoria lead by 59 runs	Junction Oval, Melbourne	2025-10-27
New Zealand vs England, 2nd ODI	New Zealand won by 5 wkts	Seddon Park, Hamilton	2025-10-29
England Women vs South Africa Women, 1st Semi-Final	South Africa Women won by 125 runs	Barsapara Cricket Stadium, Guwahati	2025-10-29
Australia vs India, 1st T20I	No result (due to rain)	Manuka Oval, Canberra	2025-10-29